



MASTG

Cryptographic Keys in Code

MASTG-TEST-0221: Broken Symmetric Encryption Algorithms

MASTG-TEST-0232: Broken Symmetric Encryption Modes

MASTG-TEST-0307: References to Asymmetric Key Pairs Used For Multiple Purposes

MASTG-TEST-0308: Runtime Use of Asymmetric Key Pairs Used For Multiple Purposes

MASTG-TEST-0309: References to Reused Initialization Vectors in Symmetric Encryption

MASTG-TEST-0310: Runtime Use of Reused

L2 ANDROID MASWE-0012 TEST



MASTG-TEST-0308: Runtime Use of Asymmetric Key Pairs Used For Multiple Purposes

Overview

This test is the dynamic counterpart to [References to Asymmetric Key Pairs Used For Multiple Purposes](#), but it focuses on intercepting cryptographic operations rather than generating keys with multiple purposes.

Some of the relevant functions to intercept are:

- `Cipher.init(int opmode, Key key, AlgorithmParameters params)` where `opmode` is one of:
 - `Cipher.ENCRYPT_MODE`
 - `Cipher.DECRYPT_MODE`
 - `Cipher.WRAP_MODE`
 - `Cipher.UNWRAP_MODE`
- `Signature.initSign(PrivateKey privateKey)`
- `Signature.initVerify(PublicKey publicKey)`

Steps

- Execute a method trace ([Method Tracing](#)) targeting all functions that use an asymmetric key to perform cryptographic operations.

Observation

The output should contain a list of all cryptographic operations together with their corresponding keys.

Evaluation

The test case fails if you find any keys used for multiple roles.

Using the output, ensure that each key (or key pair) is restricted to exactly **one** of the following groups of operations:

- Encryption/Decryption (used in `Cipher` operations with `ENCRYPT_MODE` or `DECRYPT_MODE`)
- Signing/Verification (used in `Signature` operations)
- Key Wrapping (used in `Cipher` operations with `WRAP_MODE` or `UNWRAP_MODE`)

Demos

MASTG-DEMO-0072: Runtime Use of Asymmetric Key Pairs Used For Multiple Purposes With Frida